RAJALAKSHMI ENGINEERING COLLEGE

Suriya Prakash D 251001277-2025-IT S2

Started on	Wednesday, 7 October 2026, 9:12 AM
State	Finished
Completed on	Wednesday, 7 October 2026, 9:12 AM
Time taken	21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int lowerBound(int arr[], int size, int key) {
    int left = 0, right = size;

    while (left < right) {
        int mid = left + (right - left) / 2;

        if (arr[mid] > key)
            right = mid;
        else
            left = mid + 1;
    }

    return left;
}

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int tails[n];
    int length = 0;

    for (int i = 0; i < n; i++) {
        int pos = lowerBound(tails, length, a[i]);

        tails[pos] = a[i];

        if (pos == length)
            length++;
    }

    printf("%d\n", length);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course